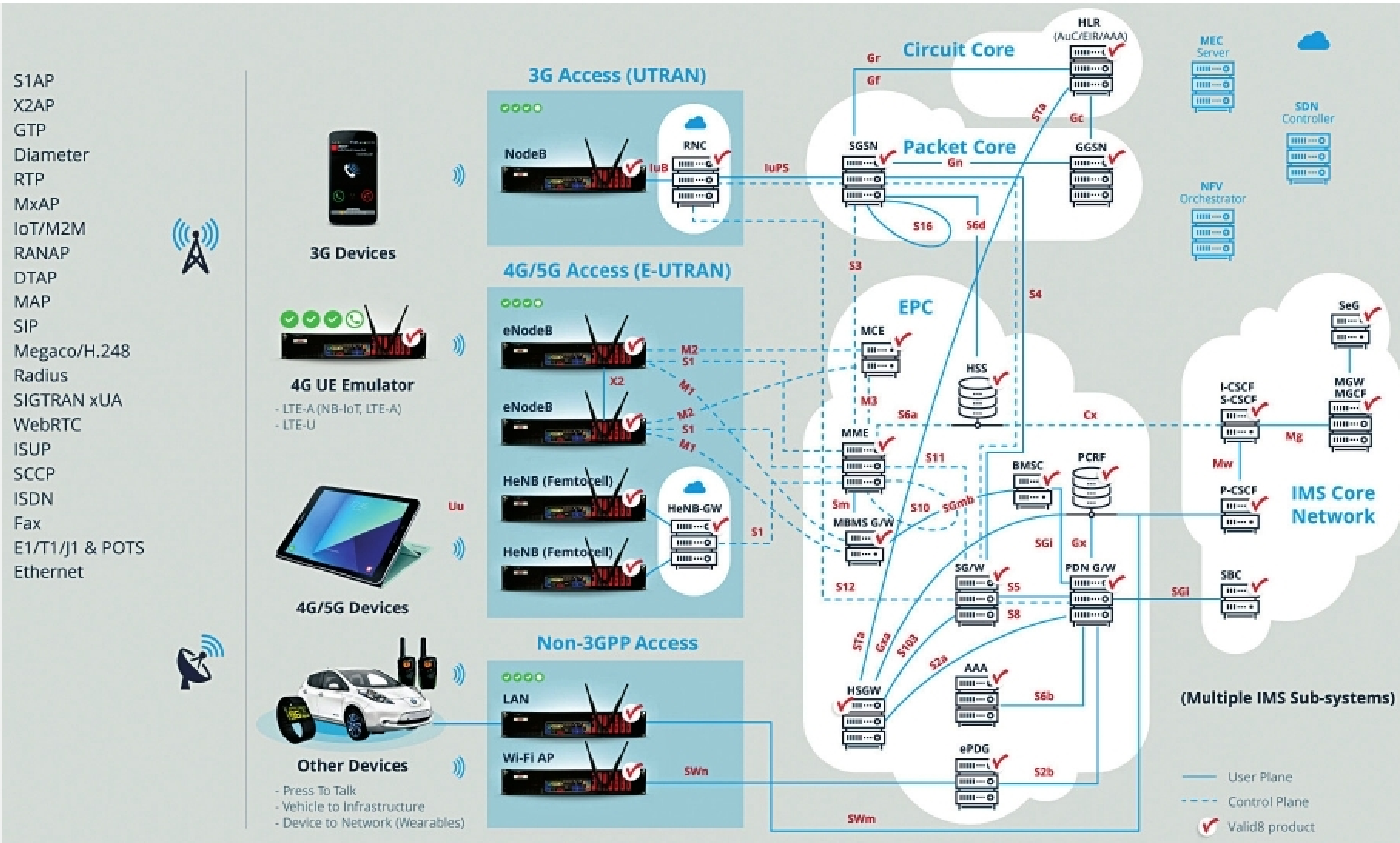
Things to Take Care of for Network Testing

There are numerous parameters that must be analysed to quantitatively and qualitatively assess a given network's performance at a time and deal with any bottlenecks that might exist. The goal is to increase network availability and minimise the time for data transfer.



(Image source: *www.aes-eu.com*)

With the push towards digitalisation and 5G, a strong network is of foremost importance. As a user, we know that this has always been the case, and the only thing that matters is that we receive the requested service as we expect. We all know how annoying it is when we want to place an order for a product, but the server crashes due to traffic. Nobody likes a website that takes forever to load, or a disconnected call, or a video that keeps buffering.

There are multiple factors why a network may not perform optimally, with one or more parameters deviating from the ideal range. By proactively identifying the potential problematic elements in a network, chances of crisis in the future are reduced. For this, understanding how the infrastructure works—both hardware and software—is crucial.

All hardware components, that is, routers, switches and cables, and servers, should be designed to meet requisite system demands based on the application. Other details like knowing the number, type and location of all devices, understanding network security architecture, operating systems, IP addresses, and wireless protocols, in advance, are highly beneficial when it comes to optimal infrastructure. Transmission Control Protocol (TCP) is the standard set of networking protocols that allows two or more computers to communicate and exchange data, and together with the Internet Protocol (IP) defines how packets of data are transmitted through a network.